DEVIN HACKATHON · TRACK 2 · MAKE THE INVISIBLE VISIBLE

TRACELESS

See what a stalker's AI sees in your photo — then make it unplaceable.

Technical & demo document · Built by a Claude Code builder lane, committed by Devin · Every number in this document was measured by the system or its benchmark scripts.

1 · The Problem

In 2019, a man in Japan found a pop idol's home from the reflection in her eyes in a photo she posted. In 2025, AI "geoguessing" went viral — tools like GeoSpy place a photo in seconds. Every social app strips EXIF location data. **Nobody strips the picture.** The pixels themselves are the leak: sign text, vegetation, architecture, road markings, and reflections carry location whether metadata exists or not.

Defenses exist mainly in research papers. We checked the product landscape and found no shipping product that detects the visual clues, removes them, and proves the removal worked. TRACELESS is that loop, end to end, on one laptop.

> **One-liner:** Upload a photo. Watch five different AIs place it. Watch the clues get erased. Watch the same AIs fail. Get the photo back on your phone, safe to post.

2 · What TRACELESS Does

1. **Clue zero.** Reads EXIF/GPS, shows it, strips it *before any attacker sees the image*. GPS is never attack input — it is only the answer key for measuring error.
2. **The attack.** Five independent geolocation models attack the pixels in parallel. Numbered clue boxes (OCR text, segmentation, faces/reflections) are found and merged with attacker-reported clues.
3. **Clue accounting.** Each clue's contribution is *measured by ablation*: erase only that clue, re-attack, record the change in 90% credible area and top-1 error.
4. **The defense.** Least-destructive fix per clue: LaMa inpainting, glyph-level text erasure, blur, reflection softening. Faces are never altered except softening eye reflections.
5. **Proof.** Re-attack with all attackers — including an adaptive attacker told the photo was edited. "Safe" means every tested attacker misses by more than the chosen radius (default 25 km). If it cannot be reached, we say so.
6. **Handback.** QR code → the judge downloads the protected photo to their own phone.

3 · Architecture

```
phone ──QR──► FastAPI backend (sessions, staged pipeline) ──state JSON──► React app
(projector)
                  │  pixels only, EXIF stripped first
                  ├─ 5 attackers in parallel (offline + cloud + street + web)
                  ├─ clue finders: OCR · dense segmentation · face/iris
                  ├─ ablation engine (measured contributions + collapse sequence)
                  ├─ defense loop (LaMa / glyph-erase / blur / soften)
                  └─ sessions, takes (PANIC replays), QR upload & download, wipe
```

Pipeline stages

STAGE	MODULE	WHAT IT DOES
exif	`exif.py`	Parse + strip all metadata; GPS kept only as answer key; verify the saved file is metadata-free.
attack	`attackers/*`	All available attackers run on pixels in parallel; each returns ranked hypotheses + rationale + a hex-grid probability distribution.
clues	`clues/*`	OCR lines, dense segmentation candidates, faces with measured iris-size gate; merged with attacker clue boxes.
ablation	`ablation.py`	Per-clue erase → re-attack → Δ credible area / Δ error. Builds the real collapse animation frame-by-frame.
defense	`defense/loop.py`	Erase top contributors → re-attack → stop when all attackers miss the radius or K rounds; honest “not reached” verdict.
card	`metrics.py`	Before/after distance, area growth, % pixels changed, SSIM/LPIPS — all computed, none typed.

The five attackers

ATTACKER	MODEL	ON/OFFLINE	ROLE
A — Cloud	Frontier vision model, strict JSON schema	Cloud	Human-level clue reading + hypotheses
B — GeoCLIP	GPS-retrieval (open source), tiled views	Offline	Primary attacker + ablation driver
Adaptive	GeoCLIP fed close-up crops	Offline	"Told the photo was edited" re-attack
C — Street	EigenPlaces → LightGlue → RANSAC	Offline	Street-level pinpoint inside coverage
Reader	OCR → cities5000 gazetteer	Offline	Reads town/shop names off signs
Sleuth	Web-research attacker	Cloud	Launched first; the ~70 s long pole

Attacker C pipeline: EigenPlaces ResNet-50+GeM descriptors → cosine top-k over a downloaded Mapillary reference set → SuperPoint keypoints + LightGlue matching → USAC-MAGSAC fundamental-matrix verification → pinpoint only if inliers clear a *measured* threshold (T = 55, measured at 95% precision within 100 m).

4 · Measured Results

Benchmark — Im2GPS3k, 300 random + 100 readable-text photos, offline attackers

33.7% → 20.3%	42% → 26%	232 → 378 km
GEOCLIP PLACEMENTS ≤25 KM (RANDOM)	≤25 KM (TEXT-RICH SET)	MEDIAN ERROR

31.3% → 22.7%	0.91	19.5%
ADAPTIVE ATTACKER ≤25 KM	SSIM AFTER DEFENSE	MEDIAN PIXELS CHANGED

Of the photos that needed defending, the 25 km target was reached on 33 of 101 (random) and 14 of 43 (text-rich). Famous, heavily photographed scenes resist — reported, not hidden. The Reader row is the pre-gazetteer-upgrade version, labelled as such.

Street level — real match, measured threshold

On the Madrid fixture, Attacker C found a *different photo of the same shop* 170 m away with 147 verified keypoint inliers against the measured threshold of 55. Street-level claims are made only inside downloaded coverage, whose size is always shown.

CLOAK — adversarial noise, honestly reported

FIXTURE	GEOCLIP ERROR BEFORE	AFTER ±2/255 NOISE	TRANSFER TO CROPS
Generic	21 km	475 km (LPIPS 0.007)	weak
Madrid (memorized scene)	0.95 km	506 km	0.43 km — still placed

White-box noise tuned to one view of one model is brittle: the adaptive attacker's close-up crops still place the photo. That is exactly why the *visible* erase is the main defense — reported as measured, including the weak transfer.

Prepared demo photo — Abergavenny bookshop

The Reader reads the town name off a sign and places it **120 m** away. After protection, every attacker misses by more than 25 km (GeoCLIP: 89 km), with 21.6% of pixels changed and **zero face pixels touched**.

Final rehearsal — 5 photos, one server, no restarts

PHOTO	ENTRY	RESULT	TIME
Abergavenny (prepared)	phone QR	120 m → 89 km, target reached	52 s
Indoor apartment	SIM	already hard to place	6 s
Madrid	drag-drop	street match found, then broken; under 1 km → honest “not reached”	36 s
Tyumen portrait	phone QR	no GPS; moved 8,846 km from the AI’s first guess	22 s
Cincinnati	SIM	honest “not reached”	64 s

5 · Truth Rules (enforced in code)

> EXIF/GPS is stripped before any attacker sees pixels; GPS is only an answer key.

> Eye reflections are claimed readable only above a measured iris-pixel threshold; below it, reflective surfaces are flagged as “possible clue” and softened as a labeled precaution.

> Every on-screen number comes from the run or the benchmark scripts. Clue contributions are measured by ablation, never invented.

> “Can’t be placed” means “not within the chosen radius, by the attackers we tested” — stated with both.

> Street-level claims only inside downloaded coverage; coverage size is always shown.

> No “locate any photo” mode. The attack is a risk report on your own photo. Exact coordinates only for public landmarks or with the owner’s OK.

> The person is never altered, except softening eye reflections. Photos stay on the laptop; wipe deletes them.

6 · The Live Demo (3 minutes)

TIME	MOMENT
0:00	Hook: the 2019 eye-reflection case, one sentence. Judge scans QR, sends a photo.
0:15	Shot 1 — the photo becomes a 3D diorama (Depth Anything V2); clue pins light up at real depths; clue zero strips.
0:35	Shot 2 — globe pulls out; hex columns collapse country → city → street as the ablation sequence plays. Rationale types in plain words.
1:00	Shots 3–4 — dive into 3D buildings; split-screen keypoint match to the street photo. "Placed X m from where you stood."
1:25	Which clues gave you away — measured contributions.
1:35	Shot 5 — particle-dissolve erasure; globe bursts back out. "Best guess now Y km off. No street match."
2:20	Shot 6 — before/after slider (on the face: untouched), the numbers, QR to the judge's phone.
2:50	"Share the moment, not the address."

Every step has a PANIC key playing its recorded take. An indoor photo routes to the honest "already hard to place" path — the failure becomes a feature.

7 · Built with Devin (the workflow itself)

This repo ran a split-brain agent workflow: **Claude Code** was the builder — architecting, writing, and self-verifying all code across multiple autonomous sessions — while **Devin** was the orchestrator and committer: issuing the brief and upgrade specs, reviewing every diff, and authoring all 12 commits on `github.com/Anandb71/traceless`. The builder logged 20+ design decisions in `DECISIONS.md` and answered the upgrade spec with `UPGRADE_PLAN.md` before touching code.

- Headless `claude -p` sessions with streamed JSON logs watched for `DEVIN_COMMIT_READY` markers; Devin committed each verified unit.
- Session continuity via `--continue`: the builder kept full context across build → upgrade → freeze → final pass.
- Self-verification: visual QA via self-screenshots, golden tests, a 5-photo rehearsal script, measured thresholds — not vibes.

8 · Stack & One-Command Start

LAYER	TECHNOLOGY
Backend	Python 3.12, FastAPI, PyTorch (CUDA), OpenCV, Pillow, NumPy
Models	GeoCLIP · EigenPlaces+LightGlue · LaMa · Depth Anything V2 · MediaPipe iris · OCR · dense segmentation
Frontend	TypeScript, React, Vite, canvas globe + diorama, synthesized audio
Ops	Makefile (`make demo` / `make bench`), QR session tokens, PANIC takes, offline-first

```
$ make demo        # ≈2 min cold, ~47 s warm — serves UI + prints URL
$ make bench       # Im2GPS3k subset benchmark → bench/results/
$ py scripts/rehearse.py   # 5-photo end-to-end rehearsal
```

All measurements in this document are traceable to files in `bench/results/` , `backend/traceless/data/` , and `takes/` . Licenses: fixture photos from Wikimedia Commons (see fixtures/ATTRIBUTION.md); Mapillary reference imagery CC BY-SA.